Effective Degrees of Freedom

from covariance penalties to the Jacobian trace

An expository companion to the LPS Tier-0 specification, gate E0.2

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This note expands one line of the E0.2 gate, the claim that $\mathrm{df} = \operatorname{tr} S = \sum_i \partial\hat{y}_i/\partial y_i$, into a short self-contained tour of what *effective degrees of freedom* means, why it is defined the way it is, and why a true mathematical identity can still be a useful software test. Everything is illustrated with a simple example and a figure, and there are exercises (with answers) at the end.

Contents

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1 The puzzle: how many parameters does a smoother use?

In ordinary least squares with p predictors, the fitted values are a linear projection of the data, $\hat{y} = Hy$, where H is the "hat matrix." Its trace counts the parameters, $\operatorname{tr} H = p$, and we call p the model's *degrees of freedom*: it is how many directions in y the fit is free to chase. Residual degrees of freedom are $n - p$, the denominator in the variance estimate.

Now fit a smoothing spline, a k-nearest-neighbour average, a local polynomial, the lasso, or a regression tree. There is no obvious "p." Yet software routinely reports things like "spline with `df` = 7.3." What is that number, and in what sense is it the right generalisation of "number of parameters"?

We want a definition of df that satisfies three things at once: it *(a)* reduces to p for OLS; *(b)* measures something operational — how much the procedure overfits; and *(c)* is computable for *any* fitting rule, linear or not. Remarkably, one quantity does all three, and it wears three equivalent faces — a covariance, an average derivative, and a matrix trace (Figure 5, later). The rest of the note builds it.

2 Optimism: why training error lies

Write the data as signal plus noise, $y_i = \mu_i + \varepsilon_i$, with the ε_i uncorrelated, mean 0, variance σ^2. A fitting procedure maps the data to fitted values, $\hat{y} = m(y)$. Two error measures matter:

- *training (apparent) error* $\mathrm{err} = \frac{1}{n}\sum_i (y_i - \hat{y}_i)^2$, the error on the very data used to fit;
- *test (prediction) error* $\mathrm{Err} = \frac{1}{n}\sum_i \mathbb{E}(y_i^* - \hat{y}_i)^2$, where $y_i^* = \mu_i + \varepsilon_i^*$ is a fresh independent observation at the same x_i and $\hat{y}$ is held fixed from the original data.

Training error is optimistic: the fit has already seen y, so it explains part of the noise. The gap is exactly a covariance.

Theorem 1 (Optimism)**.** $\mathbb{E}[\mathrm{Err}] - \mathbb{E}[\mathrm{err}] \;=\; \dfrac{2}{n}\displaystyle\sum_{i=1}^{n} \mathrm{Cov}(\hat{y}_i, y_i).$

Proof. Work point by point. Because y_i^* is an independent copy of y_i, it has the same first two moments, and y_i^* is independent of $\hat{y}_i$ (which is a function of the original y). Expanding the squares,

$$\mathbb{E}[(y_i^* - \hat{y}_i)^2] - \mathbb{E}[(y_i - \hat{y}_i)^2] = \underbrace{\left(\mathbb{E}[y_i^{*2}] - \mathbb{E}[y_i^2]\right)}_{=\,0} - 2\left(\mathbb{E}[y_i^*\hat{y}_i] - \mathbb{E}[y_i\hat{y}_i]\right).$$

Independence gives $\mathbb{E}[y_i^*\hat{y}_i] = \mathbb{E}[y_i^*]\,\mathbb{E}[\hat{y}_i] = \mu_i\mathbb{E}[\hat{y}_i]$, while $\mathbb{E}[y_i\hat{y}_i] = \mathrm{Cov}(y_i, \hat{y}_i) + \mu_i\mathbb{E}[\hat{y}_i]$. The difference is $-\mathrm{Cov}(y_i, \hat{y}_i)$, so the bracket equals $2\,\mathrm{Cov}(\hat{y}_i, y_i)$. Average over i. □

So $\sum_i \mathrm{Cov}(\hat{y}_i, y_i)$ *is* the amount of overfitting — how much the fit's values move *with* their own observations. Normalising by the noise level turns it into a pure count.

Key idea. Efron's definition of effective degrees of freedom is $\mathrm{df} \;=\; \dfrac{1}{\sigma^2}\displaystyle\sum_{i=1}^{n}\mathrm{Cov}(\hat{y}_i, y_i)$. It is the covariance penalty that corrects training error: $\mathbb{E}[\mathrm{Err}] \approx \mathbb{E}[\mathrm{err}] + \frac{2\sigma^2}{n}\,\mathrm{df}$ (Mallows' C_p).

Example 1 (OLS recovers p)**.** For $\hat{y} = Hy$, $\mathrm{Cov}(\hat{y}_i, y_i) = \mathrm{Cov}\left(\sum_j H_{ij}y_j,\, y_i\right) = H_{ii}\sigma^2$, since distinct observations are uncorrelated. Hence $\mathrm{df} = \sigma^{-2}\sum_i H_{ii}\sigma^2 = \mathrm{tr}\,H = p$. A straight-line fit ($p = 2$: intercept and slope) has $\mathrm{df} = 2$ and optimism $4\sigma^2/n$.

Figure 1 shows the mechanism directly: as model complexity grows, training error falls monotonically while test error turns up, and the gap between them widens in exact proportion to df.

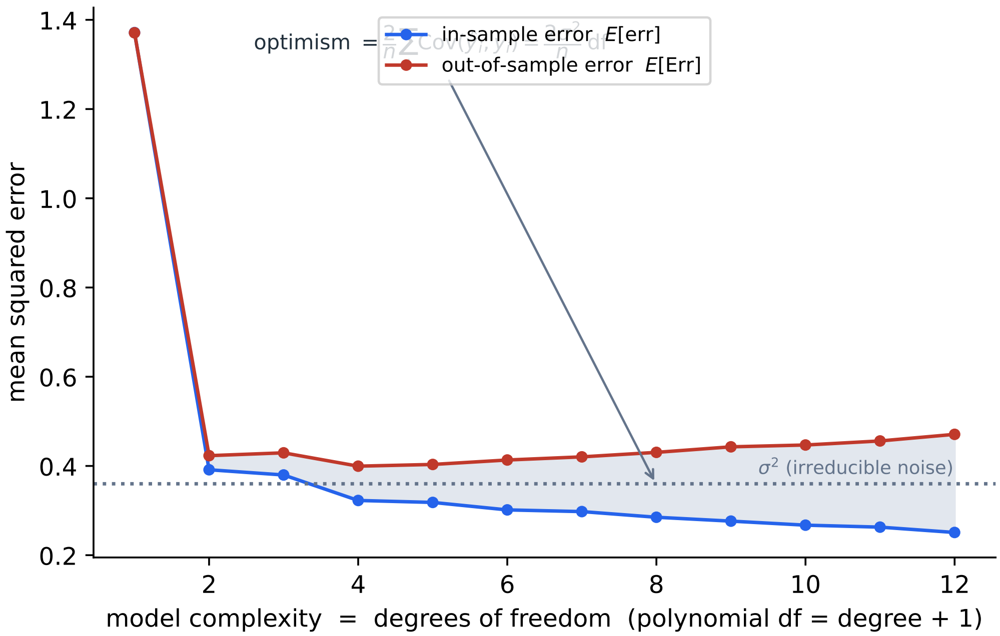

Figure 1: Polynomials of increasing degree fit to noisy data (averaged over 500 replications). In-sample error keeps dropping, but out-of-sample error is U-shaped; the shaded gap is the optimism $\frac{2}{n}\sum_i \mathrm{Cov}(\hat{y}_i, y_i) = \frac{2\sigma^2}{n}\,\mathrm{df}$. Degrees of freedom is precisely the knob that sets how far training error understates the truth.

3 The smoother matrix and self-sensitivity

A *linear smoother* writes the fit as $\hat{y} = Sy$ for a matrix S that depends on the design X (positions, bandwidth, degree, kernel) but *not* on y. The diagonal entry

$$S_{ii} \;=\; \frac{\partial \hat{y}_i}{\partial y_i}$$

is the *self-sensitivity* (or leverage) of point i: nudge the single observation y_i and watch how much its own fitted value follows. A fit that chases the data has $S_{ii} \approx 1$; a heavy smoother that averages many points has $S_{ii} \approx 0$ (Figure 2).

The covariance definition collapses to a trace here, and — worth noting — with no distributional assumption beyond uncorrelated, equal-variance errors:

$$\mathrm{Cov}(\hat{y}_i, y_i) = \mathrm{Cov}\left(\textstyle\sum_j S_{ij} y_j,\; y_i\right) = S_{ii}\,\sigma^2 \quad\Longrightarrow\quad \mathrm{df} = \textstyle\sum_i S_{ii} = \mathrm{tr}\,S.$$

Key idea. For any linear smoother $\hat{y} = Sy$, $\mathrm{df} = \mathrm{tr}\,S$, the sum of the self-sensitivities. This is the form most often quoted for splines and kernel smoothers.

Example 2 (k-nearest-neighbour average)**.** If $\hat{y}_i$ is the average of the k observations nearest x_i (counting point i itself), then the weight on y_i is $1/k$, so $S_{ii} = 1/k$ and $\mathrm{df} = \mathrm{tr}\,S = n/k$. With $k = 1$ the smoother interpolates, $\mathrm{df} = n$; with $k = n$ it is the global mean, $\mathrm{df} = 1$. Degrees of freedom slides continuously between these as k changes.

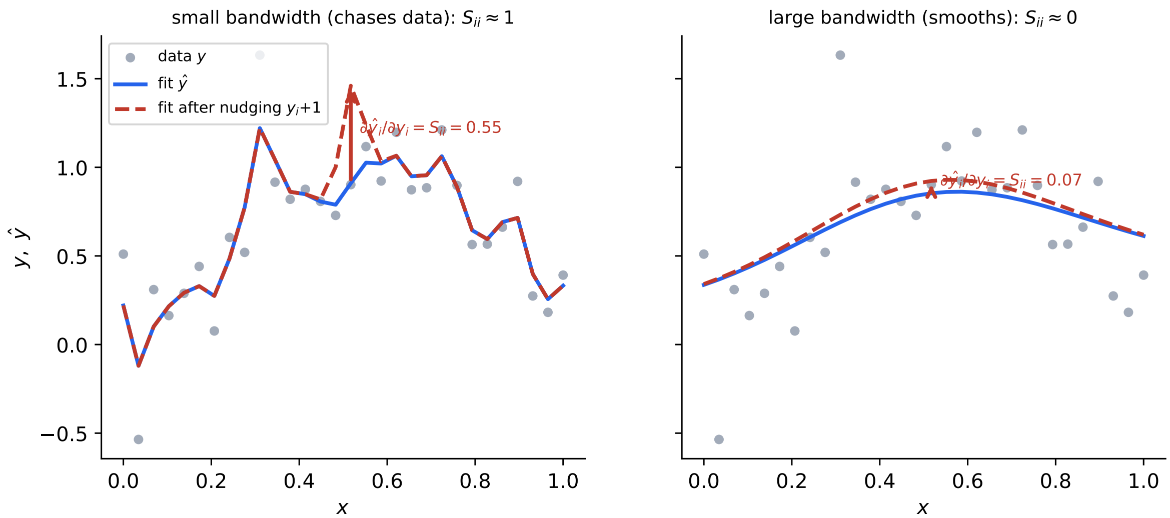

Figure 2: Self-sensitivity made visible. The same point y_i is nudged up by 1 (solid → dashed). With a small bandwidth (left) the fit tracks the point almost one-for-one, $S_{ii} \approx 1$; with a large bandwidth (right) the fit barely moves, $S_{ii} \approx 0$. The vertical red arrow is the realised change $\hat{y}_i$, which equals S_{ii} because the nudge has size 1.

4 Stein's lemma

To extend "df $= \operatorname{tr} S$" to *nonlinear* fits we need a bridge from the covariance form to a derivative. That bridge is Stein's lemma, a small but powerful consequence of integration by parts against the Gaussian density.

Lemma 1 (Stein, 1981)**.** *If $Z \sim N(\mu, \sigma^2)$ and g is absolutely continuous with $\mathbb{E}|g'(Z)| < \infty$, then* $\operatorname{Cov}(Z, g(Z)) = \sigma^2\, \mathbb{E}[g'(Z)]$.

Proof. The normal density φ satisfies $\varphi'(z) = -\frac{z-\mu}{\sigma^2}\varphi(z)$, i.e. $(z-\mu)\varphi(z) = -\sigma^2\varphi'(z)$ (Figure 4, left). Then

$$\mathbb{E}[(Z-\mu)g(Z)] = \int (z-\mu)g(z)\varphi(z)\,dz = -\sigma^2 \int g(z)\varphi'(z)\,dz = \sigma^2 \int g'(z)\varphi(z)\,dz = \sigma^2\mathbb{E}[g'(Z)],$$

integrating by parts (the boundary terms vanish under the stated conditions). □

The multivariate version, applied coordinatewise: if $y \sim N(\mu, \sigma^2 I_n)$ and $h : \mathbb{R}^n \to \mathbb{R}$ is weakly differentiable with integrable partials, then $\operatorname{Cov}(y_i, h(y)) = \sigma^2\, \mathbb{E}[\partial h/\partial y_i]$.

Example 3 (two quick checks)**.** For $g(z) = z$: $\operatorname{Cov}(Z, Z) = \sigma^2$ and $\sigma^2\mathbb{E}[g'] = \sigma^2$. For $g(z) = z^2$ with $Z \sim N(\mu, \sigma^2)$: $\operatorname{Cov}(Z, Z^2) = 2\mu\sigma^2$, and $\sigma^2\mathbb{E}[2Z] = 2\mu\sigma^2$. The soft-threshold g of Figure 4 has $g' = \mathbf{1}\{|z| > \lambda\}$, so $\operatorname{Cov}(Z, g(Z)) = \sigma^2 \Pr(|Z| > \lambda)$ — a preview of the lasso.

5 Putting it together: covariance = expected divergence

Apply the multivariate lemma with $h(y) = \hat{y}_i = m_i(y)$, the i-th fitted value seen as a function of the whole data vector:

$$\operatorname{Cov}(\hat{y}_i, y_i) = \sigma^2\, \mathbb{E}\Big[\frac{\partial \hat{y}_i}{\partial y_i}\Big].$$

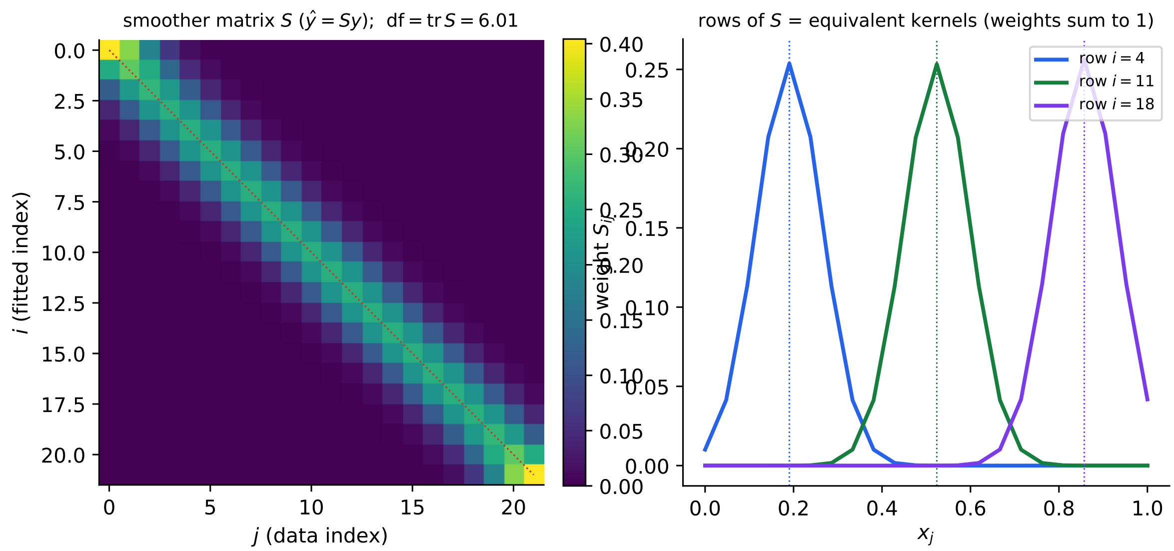

Figure 3: Left: the smoother matrix S of a kernel smoother; the dotted line is the diagonal, and $\mathrm{df} = \operatorname{tr} S$ sums those diagonal entries. Right: individual rows of S are the "equivalent kernels" — the weights each fitted value places on the data (they sum to 1).

Summing over i and dividing by σ^2,

$$\boxed{\mathrm{df} \;=\; \frac{1}{\sigma^2}\sum_i \operatorname{Cov}(\hat{y}_i, y_i) \;=\; \mathbb{E}\Big[\sum_i \frac{\partial \hat{y}_i}{\partial y_i}\Big] \;=\; \mathbb{E}\big[\operatorname{tr} J(y)\big]}$$

where $J(y) = \partial\hat{y}/\partial y$ is the Jacobian of the fitting map, with entries $J_{ij} = \partial\hat{y}_i/\partial y_j$. The sum of its diagonal, $\sum_i \partial\hat{y}_i/\partial y_i$, is the *divergence* of $y \mapsto \hat{y}$. So degrees of freedom is the expected divergence of the fit — "Stein's unbiased degrees of freedom," the quantity inside the SURE risk estimate.

Two remarks pin down the logical status. *First*, for a linear smoother $J(y) = S$ for every y, so the expectation is vacuous and $\mathrm{df} = \operatorname{tr} S$ exactly — no Gaussianity needed. *Second*, for a general fit the Jacobian depends on y, and the *sample* divergence $\operatorname{tr} J(y)$ evaluated at the observed data is an *unbiased estimate* of df. When no formula exists, one estimates it by finite differences or random perturbations of y (Ye's "generalized degrees of freedom"). This sample divergence is exactly what the E0.2 bump-trace computes.

6 A gallery of degrees of freedom

- **OLS:** $\mathrm{df} = \operatorname{tr} H = p$ (an integer; the anchor case).
- **Ridge regression:** $\mathrm{df}(\lambda) = \sum_j d_j^2/(d_j^2 + \lambda)$, where d_j are the singular values of the design. It decreases smoothly from p (at $\lambda = 0$) to 0 (as $\lambda \to \infty$); see Figure 6. Degrees of freedom is generally *not an integer*.
- **Smoothing spline:** $\mathrm{df} = \operatorname{tr} S(\lambda)$; practitioners often choose and report df rather than the opaque penalty λ.
- **k-NN:** $\mathrm{df} = n/k$.

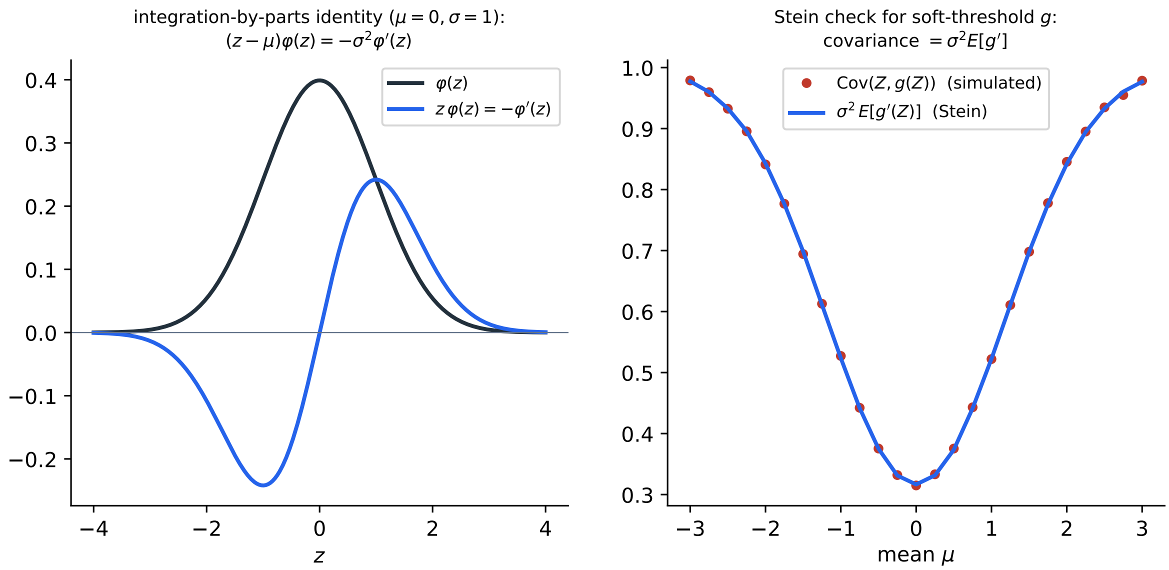

Figure 4: Left: the integration-by-parts identity behind Stein's lemma, $(z-\mu)\varphi(z) = -\sigma^2\varphi'(z)$. Right: a numerical check for the soft-threshold $g(z) = \mathrm{sign}(z)(|z| - \lambda)_+$, whose derivative is the indicator $\mathbf{1}\{|z| > \lambda\}$: the simulated covariance matches $\sigma^2\mathbb{E}[g']$ across the mean μ.

- **Lasso / soft-threshold:** df $= \mathbb{E}[\,\#\{\text{nonzero coefficients}\}\,]$ (Zou–Hastie–Tibshirani). In the orthonormal case this follows from the soft-threshold derivative of the previous section: each coordinate contributes $\mathbf{1}\{|z_j| > \lambda\}$. Degrees of freedom literally counts the variables the procedure *effectively selects* — but as an expectation that already accounts for the adaptivity of the selection.

- **Interpolation:** $\hat{y} = y$ gives $S = I$, df $= \mathrm{tr}\, I = n$ — maximal complexity, zero training error, and (Exercise 6) useless predictions.

Key idea. "This definition applies to any fitter, linear or not." The self-sensitivity / divergence definition needs no parameter count, no linear structure, and no integer — only that the fit responds to perturbing the data. So it assigns a meaningful, comparable df to splines, k-NN, the lasso, trees, even neural nets. The classical "df = number of parameters" is the special case where $\hat{y} = Hy$ and $\mathrm{tr}\, H = p$ happens to be an integer.

7 Back to the test: E0.2 as software falsification

For the *ideal* linear operator S, the equality df $= \mathrm{tr}\, S = \sum_i \partial\hat{y}_i/\partial y_i$ is a theorem — it cannot be empirically false. So what does E0.2 test? Not the theorem; the *code*. There are two objects: the linear-algebra ideal S, and the compiled function `fit.lps` that may or may not actually realise it. The test computes the two sides by two *independent* routes and checks they agree:

- **Route A** — reconstruct S column by column from $S_{\cdot j} = \texttt{fit}(e_j)$ (feed unit-basis responses), then take $\mathrm{tr}\, S$.
- **Route B** — finite-difference the self-sensitivities around a *generic* random y: bump y_i by $\varepsilon = 1$ and sum $\texttt{fit}(y + e_i)_i - \texttt{fit}(y)_i$.

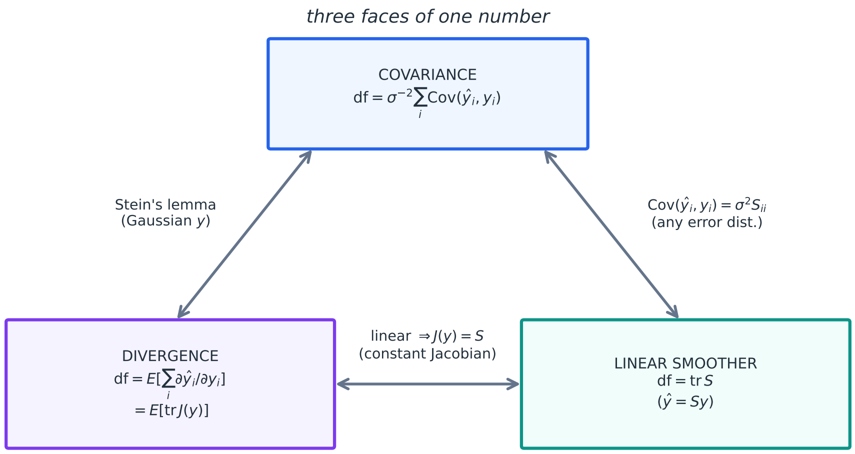

Figure 5: One number, three faces. Efron's covariance form and the divergence/Jacobian-trace form are linked by Stein's lemma (and so require Gaussian y); both collapse to $\operatorname{tr} S$ for a linear smoother, where the Jacobian $J(y) = S$ is constant. The covariance-to-$\operatorname{tr} S$ edge needs only uncorrelated, equal-variance errors — no normality.

If the implemented map is genuinely linear and y-independent, the two routes agree exactly (the theorem), and the step $\varepsilon = 1$ is exact because for a linear map the secant equals the derivative with no truncation error. If instead there is any hidden y-dependence, the Jacobian varies with the operating point, the two routes probe different points ($\{e_j\}$ near the origin versus a generic y), and they *disagree* (Figure 7).

Real culprits for such hidden y-dependence: a bandwidth chosen from the spread of y; a ridge that activates only when a y-dependent condition number exceeds a cap; probability clipping; or an "unstable-action" fallback keyed to the magnitudes of y. Any of these would silently break every downstream variance, df, and CV-shortcut formula that assumes a fixed S.

Key idea. The theorem is the *oracle*; the test is a *falsification* of the software. A correct implementation is forced to satisfy a relationship a buggy (nonlinear, y-dependent, or mis-accounted) one will violate — which is why a "trivially true" identity makes a sharp unit test.

Exercises

Exercise 1. A k-NN running-mean smoother sets $\hat{y}_i$ to the average of the k observations nearest x_i, counting i itself. Show $S_{ii} = 1/k$ and $\mathrm{df} = n/k$. What are the two extremes $k = 1$ and $k = n$?

Exercise 2. On $n = 6$ ordered points, the width-3 moving average sets $\hat{y}_i = (y_{i-1} + y_i + y_{i+1})/3$ for interior i, and at the ends $\hat{y}_1 = (y_1 + y_2)/2$, $\hat{y}_6 = (y_5 + y_6)/2$. Compute $\operatorname{tr} S$.

Exercise 3. Let $Z \sim N(0, 1)$ and $g(z) = z^3$. Compute $\mathrm{Cov}(Z, Z^3)$ and $\sigma^2\mathbb{E}[g'(Z)]$ directly and check Stein's lemma.

Exercise 4. For ridge, $\mathrm{df}(\lambda) = \sum_j d_j^2/(d_j^2 + \lambda)$. Show $\mathrm{df}(\lambda)$ is strictly decreasing in λ, with limits p and 0.

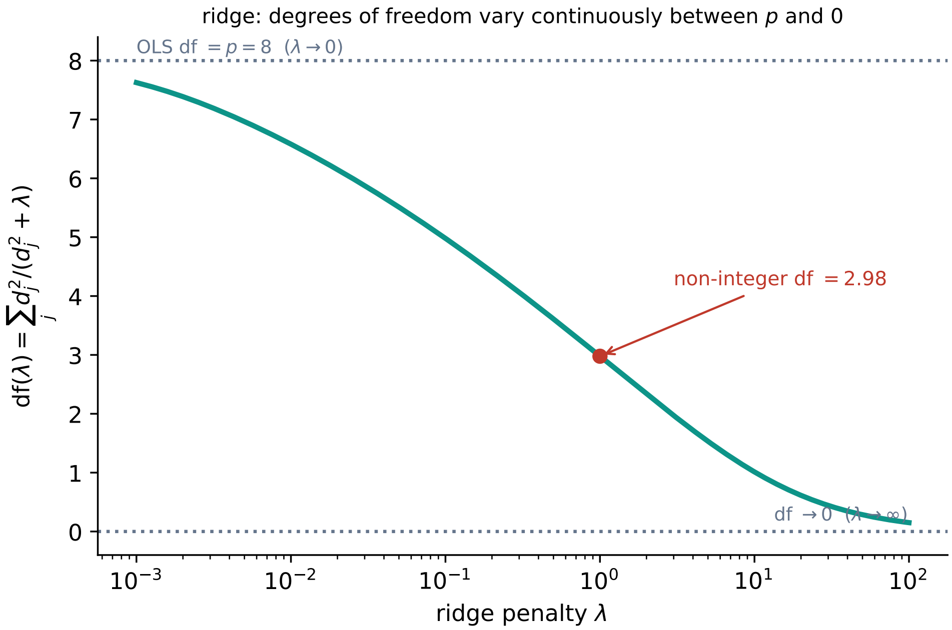

Figure 6: Ridge degrees of freedom $\mathrm{df}(\lambda) = \sum_j d_j^2/(d_j^2+\lambda)$ vary continuously between the parameter count p and 0. "Effective" degrees of freedom is generally a real number, not a count.

Exercise 5. In an orthonormal design the lasso gives $\hat{\beta}_j = \mathrm{sign}(z_j)(|z_j|-\lambda)_+$ with $z = X^\top y$. Using the divergence definition, show $\mathrm{df} = \mathbb{E}[\,\#\{j : |z_j| > \lambda\}\,]$, the expected number of selected variables.

Exercise 6. A fit interpolates the data, $\hat{y} = y$. What is df? Using Theorem 1, what is the test error, and why is a zero training error misleading?

Exercise 7. In the E0.2 test, suppose route A ($\mathrm{tr}\, S$ from `fit`(e_j)) and route B (finite-difference trace around a random y) disagree well beyond rounding. What does this tell you about the implementation, and give one concrete code bug that would cause it?

Answers

1. $\hat{y}_i = \frac{1}{k}\sum_{j \in N_i} y_j$ with $i \in N_i$, so the coefficient on y_i is $1/k$, i.e. $S_{ii} = 1/k$. Then $\mathrm{df} = \mathrm{tr}\, S = \sum_{i=1}^n \frac{1}{k} = n/k$. $k = 1$: $\hat{y} = y$, $\mathrm{df} = n$ (interpolation); $k = n$: global mean, $\mathrm{df} = 1$.

2. Interior indices $i = 2, 3, 4, 5$ each contribute $S_{ii} = 1/3$; the two ends contribute $S_{11} = S_{66} = 1/2$. So $\mathrm{tr}\, S = 4 \cdot \frac{1}{3} + 2 \cdot \frac{1}{2} = \frac{4}{3} + 1 = \frac{7}{3} \approx 2.33$. (A width-3 average "spends" about 2.3 parameters on 6 points.)

3. $\mathrm{Cov}(Z, Z^3) = \mathbb{E}[Z^4] - \mathbb{E}[Z]\mathbb{E}[Z^3] = 3 - 0 = 3$. And $\sigma^2\mathbb{E}[g'(Z)] = 1 \cdot \mathbb{E}[3Z^2] = 3\mathbb{E}[Z^2] = 3$. They agree.

4. Each term $t_j(\lambda) = d_j^2/(d_j^2 + \lambda)$ has $t_j'(\lambda) = -d_j^2/(d_j^2 + \lambda)^2 < 0$, so df is strictly decreasing. As $\lambda \to 0$, $t_j \to 1$ and $\mathrm{df} \to p$; as $\lambda \to \infty$, $t_j \to 0$ and $\mathrm{df} \to 0$. Thus df ranges continuously over $(0, p)$.

5. With an orthonormal design the coordinates decouple and $\partial\hat{\beta}_j/\partial z_j = \mathbf{1}\{|z_j| > \lambda\}$ (the soft-threshold is flat inside $[-\lambda, \lambda]$ and unit-slope outside). The fitted values inherit the same diagonal Jacobian, so $\sum_i \partial\hat{y}_i/\partial y_i = \#\{j : |z_j| > \lambda\}$, and taking expectations, $\mathrm{df} = \mathbb{E}[\#\{|z_j| > \lambda\}]$ — the

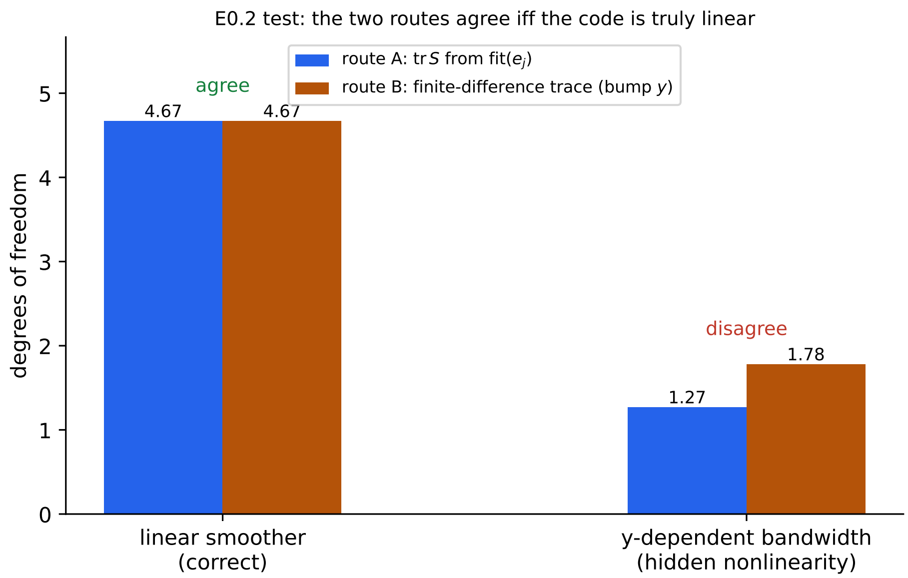

Figure 7: The two routes on a genuine linear smoother (left) agree to rounding; on a "smoother" whose bandwidth secretly depends on y (right) they disagree, because its Jacobian is not constant. The gap is a falsification signal: the code is not the linear operator it claims to be.

expected size of the active set. (Note it is an expectation: the realised active-set size is the *unbiased estimate* of df.)

6. $S = I$, so $\mathrm{df} = \mathrm{tr}\, I = n$. Every $\partial \hat{y}_i / \partial y_i = 1$. Training error is 0, but by Theorem 1 the optimism is $\frac{2\sigma^2}{n} \cdot n = 2\sigma^2$, so $\mathbb{E}[\mathrm{Err}] \approx 2\sigma^2$ — *twice* the irreducible noise. A perfect in-sample fit with the worst possible generalisation: the cautionary tale of $\mathrm{df} = n$.

7. Disagreement means the implemented Jacobian depends on the operating point — the map is not linear, or S secretly depends on y. A concrete bug: the bandwidth (or support radius) is computed from the data spread of y, so two different response vectors get different weights; then `fit`(e_j) (probing near 0) and the bumps around a generic y see different smoothers. Any y-dependent ridge trigger, clipping, or fallback does the same. The consequence is that $\mathrm{df} = \mathrm{tr}\, S$ is no longer well defined and every variance / CV-shortcut built on it is biased — exactly the silent failure E0.2 is designed to catch.

References

- B. Efron (1986). How biased is the apparent error rate of a prediction rule? *JASA* 81, 461–470. (Optimism and the covariance penalty.)
- B. Efron (2004). The estimation of prediction error: covariance penalties and cross-validation. *JASA* 99, 619–632.
- C. Stein (1981). Estimation of the mean of a multivariate normal distribution. *Ann. Statist.* 9, 1135–1151. (Stein's lemma and SURE.)
- J. Ye (1998). On measuring and correcting the effects of data mining and model selection. *JASA* 93, 120–131. (Generalized degrees of freedom.)
- H. Zou, T. Hastie, R. Tibshirani (2007). On the degrees of freedom of the lasso. *Ann. Statist.* 35, 2173–2192.
- R. J. Tibshirani, J. Taylor (2012). Degrees of freedom in lasso problems. *Ann. Statist.* 40, 1198–1232.

- T. Hastie, R. Tibshirani, J. Friedman. *The Elements of Statistical Learning*, §7.6 (effective number of parameters) and §5.4 (smoother matrices).